Variational inference

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Overview

Today, we cover:

- Variational inference
- Example with linear regression, mixture modeling

Resources:

- Variational Inference: A Review for Statisticians by David Blei

Announcements:

- Final Project proposal due 4/1 at 10AM
- Homework 4 and 5 will be combined, posted by Friday

For class tomorrow

- Install VPN and try logging into the cluster
- Try getting into my partition

Final project proposal

Due **April 1**!

Your final project proposal should be one page or less and clearly outline your plans for your final project. This document will help ensure your project is well-defined and feasible.

Your proposal should include the following elements:

1. **Project Title**
2. **Objective**: Clearly state the main goal of your project. What problem are you addressing, or what question are you trying to answer?
3. **Background & Motivation**: Provide brief context on why this project is important or interesting. Why did you choose this topic?
4. **Approach**: Explain how you plan to complete the project, specifically highlighting concepts, techniques, or tools from class that you will use.

Joint posterior distribution

In Bayesian settings our goal is to learn about the joint posterior distribution, i.e.

$$p(\beta, \sigma_y^2 | y, X, hyperparameters)$$

This can be hard!

- Often the joint posterior is analytically intractable
- MCMC: sample from it by constructing a Markov chain whose stationary distribution is the posterior
    - Can be very slow to converge
    - High computational cost

Variational inference motivation

Our goal is still to learn about the joint posterior distribution

$$p(\beta, \sigma_y^2 | y, X, hyperparameters)$$

- Variational inference turns this from a sampling problem into an **optimization** problem
- Uses a surrogate distribution to approximate the joint density of interest and maximizes the surrogate instead
- Scales much better to large datasets

Two-part Gaussian mixture model

- $Y_1, \dots, Y_n$ are sampled independently from a mixture of two Normal distributions with density

$$p(y|\theta) = \lambda \mathcal{N}(y|\mu_1, \sigma_1^2) + (1-\lambda)\mathcal{N}(y|\mu_2, \sigma_2^2)$$

- $\theta = (\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \lambda)$
- $c_1, \dots, c_n$: labels identifying which observation came from which population
  - $c_i = 1$ if Y_i from $\mathcal{N}(y|\mu_1, \sigma_1^2)$; $c_i = 0$ otherwise

$$c_i \sim Bernoulli(\lambda)$$

Two-part Gaussian mixture model

Joint density of observed and missing data (i.e. complete data density) is then

$$p(y, c|\theta) = \left[\lambda \mathcal{N}(y|\mu_1, \sigma_1^2)\right]^c \left[(1-\lambda)\mathcal{N}(y|\mu_2, \sigma_2^2)\right]^{1-c}$$

Bayesian Gaussian K-mixture model

Data generating model is

$$\mu \sim N(0, \sigma^2)$$

$$c_i \sim Cat(\lambda); \sum_{k=1}^{K} \lambda_k = 1$$

$$y_i | c_i, \mu \sim N(\mu_{c_i}, 1)$$

Bayesian Gaussian mixture model

Posterior PDF (ignoring hyperparameters) is

$$p(\mu, \mathbf{c}|\mathbf{y}) = \frac{\prod_k p(\mu_k) \prod_i p(c_i) p(y_i|c_i, \mu)}{\int_\mu \sum_c \prod_k p(\mu_k) \prod_i p(c_i) p(y_i|c_i, \mu)}$$

- Because of denominator integral cannot be computed easily analytically

Bayesian linear regression

$$y = X\beta + \epsilon$$

$$\epsilon \sim N(0, \sigma_y^2)$$
$$\beta \sim N(0, \sigma_b^2)$$
$$\sigma_y^2 \sim IG(A, B)$$

Joint posterior we want to estimate is

$$p(\beta, \sigma_y^2 | y, X, h) = \frac{p(\beta)p(\sigma^2)\prod_i p(y|\beta, \sigma_y^2)}{\int_\beta \int_{\sigma_y^2} p(\beta)p(\sigma^2)\prod_i p(y|\beta, \sigma_y^2) d\beta d\sigma_y^2}$$

General setup for variational inference

We have a posterior distribution / other probability distribution with an intractiable integral that makes it challen ging to evaluate.

- **MCMC**: sample from it
- **VI**: turn it into a tractable optimization problem

General setup for variational inference

Consider a joint density of latent variables $\mathbf{z} = z_{1:m}$ and observations $\mathbf{y} = y_{1:n}$:

$$p(\mathbf{z}, \mathbf{y}) = p(\mathbf{z})p(\mathbf{y}|\mathbf{z})$$

Latent variables here could mean latent in the E-M algorithm sense, or just parameters we want to estimate.

- Bayesian mixture models:
- Bayesian linear regression:

Main idea behind variational Bayes

Choose a family of approximate densities, $\mathcal{Q}$ over the latent variables z. Then, find a specific member of that family $q(z|\nu) \in \mathcal{Q}$ that minimizes the **Kullback-Leibler (KL) divergence** to the exact posterior, $p(z|y)$. Here, ν is a variational factor we find the optimal value of.

$$KL(q \parallel p) = \int_z q(z) \log \frac{q(z)}{p(z|y)} dz = E_q \left[\log \frac{q(z)}{p(z|y)} \right]$$

- Goal is to minimize $KL(q \parallel p)$
- Then, we approximate the posterior with the optimized member of the family $q^*(\cdot)$

The evidence lower bound

- Can't actually minimize $KL(q \parallel p)$ directly, but can optimize a function that is equal to it up to a constant.
- Called **evidence lower bound (ELBO)**
- In this setting, "evidence" is a term for the marginal likelihood of observed data, $p(y)$.

The evidence lower bound

$$
\begin{aligned}
KL(q \parallel p) &= E_q\left[\log \frac{q(z)}{p(z|y)}\right] \\
&= E[\log q(z)] - E[\log p(z,y)] + \log p(y)
\end{aligned}
$$

ELBO is

$$
ELBO(q) = E[\log p(z,y)] - E[\log q(z)]
$$

- Maximizing $ELBO(q)$ is equivalent to minimizing $KL(q \parallel p)$.

The evidence lower bound

Why is it called the evidence lower bound? Serves as a lower bound for $p(y)$ by Jensen's Inequality:

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Why is it called the evidence lower bound? Serves as a lower bound for $p(y)$ by Jensen's Inequality:

$$
\begin{aligned}
\log p(y) &= \log \int_z p(y, z) dz \\
&= \log \int_z p(y, z) \frac{q(z)}{q(z)} \\
&= \log \left(E_q \left[\frac{p(y, z)}{q(z)} \right] \right) \\
&\geq E[\log p(y, z)] - E[\log q(z)]
\end{aligned}
$$

The evidence lower bound

- Optimize this over densities $q(z)$ to find optimal approximation
- In variational inference, we find settings of the variational density that maximize the ELBO, which is equivalent to minimizing the KL divergence.
- We choose a family of variational distributions such that these two expectations can be computed

- How to choose the family of densities?

Mean field variational inference

Mean field makes the strong simplifying assumption that the latent variables are independent in the approximation:

$$q(\mathbf{z}) = \prod_{j=1}^{m} q_j(z_j)$$

- The variational family is not a model of the observed data! Instead, it is the ELBO and the corresponding KL minimization problem which connects the fitted variational density to the data.

Mean field variational inference

In principle, q_j can take on any parametric form appropriate to the corresponding variable z_j.

- For example, continuous variable might have a Gaussian factor
- Categorical variable might have a categorical factor

- Allows for optimization of each z_j independently

Coordinate Ascent Variational Inference (CAVI)

- Iterative algorithm for solving this optimization problem
- CAVI iteratively optimizes each factor of the mean-field variational density while holding the others fixed
- Climbs ELBO to a **local** optimum
- *Complete conditional* of jth latent variable z_j is $p(z_j|\mathbf{z}_{-j}, \mathbf{y})$.

Optimal $q_j(z_j)$ is

$$q_j^*(z_j) = \exp\left\{E_{-j}\left[\log p(z_j|\mathbf{z}_{-j}, \mathbf{y})\right]\right\}$$

where expectation is taken W.R.T. $q_{-j}(\mathbf{z}_{-j}) = \prod_{l \neq j} q_l(z_l)$.

Coordinate Ascent Variational Inference (CAVI)

Algorithm 1: Coordinate ascent variational inference (CAVI)

Input: A model $p(\mathbf{x}, \mathbf{z})$, a data set $\mathbf{x}$
Output: A variational density $q(\mathbf{z}) = \prod_{j=1}^{m} q_j(z_j)$
Initialize: Variational factors $q_j(z_j)$
while *the ELBO has not converged* **do**
  for $j \in \{1, \ldots, m\}$ **do**
    Set $q_j(z_j) \propto \exp\{\mathbb{E}_{-j}[\log p(z_j \mid \mathbf{z}_{-j}, \mathbf{x})]\}$
  end
  Compute $\textsc{elbo}(q) = \mathbb{E}[\log p(\mathbf{z}, \mathbf{x})] + \mathbb{E}[\log q(\mathbf{z})]$
end
return $q(\mathbf{z})$

Derivation of CAVI updates

Goal is to find $q(z)$ that maximizes ELBO. First, some useful results.

- by probability chain rule:

$$p(\mathbf{y}, z) = p(\mathbf{y}) \prod_{j=1}^{m} p(z_j | \mathbf{z}_{-j}, \mathbf{y})$$

- using mean field approximation $q(\mathbf{z}) = \prod_{j=1}^{m} q_j(z_j)$:

$$E_q[\log q(\mathbf{z})] = \sum_{j=1}^{m} E_{q_j}[\log q_j(z_j)]$$

Bayesian linear regression

$$y = X\beta + \epsilon$$

$$\epsilon \sim N(0, \sigma_y^2)$$
$$\beta \sim N(0, \sigma_b^2)$$
$$\sigma_y^2 \sim IG(A, B)$$

- Posterior is $p(\beta, \sigma_y^2 | y, x)$
- Approximate with q

Bayesian linear regression

From mean field assumption,

$$q(\beta, \sigma_y^2) = q(\beta)q(\sigma_y^2)$$

$$ELBO(q) = E_q[\log p(y, \beta, \sigma_y^2)] - E_q[\log q(\beta)] - E_q[\log q(\sigma_y^2)]$$

$$\log q^*(\beta) \propto E_{q(\sigma_y^2)}[\log p(y, \beta, \sigma_y^2)]$$
$$\log q^*(\sigma^2) \propto E_{q(\beta)}[\log p(y, \beta, \sigma_y^2)]$$

Solve these expectations to get updates for β, σ^2

Bayesian linear regression

Update for β is

$$q^*(\beta) = N(m, S)$$

$$S = (\frac{1}{\sigma_y^2} x^T x + \frac{1}{\sigma_b^2} I)^{-1}$$

$$m = S(\frac{1}{\sigma_y^2} x^T y)$$

Bayesian linear regression

Update for σ_y^2 is

$$q^*(\sigma_y^2) = IG(A_{post}, B_{post})$$

$$A_{post} = A_0 + n/2$$

$$B_{post} = B_0 + 1/2 E_{q(\beta)}[(y - x\beta)^2] = B_0 + 1/2\left((y - xm)^2 + tr(x^t x S)\right)$$

BLR CAVI Algorithnm:

1. Initialize σ_y^2
2. Repeat until ELBO converges:

- Update q( ) by computing S and m using current $1/\sigma_y^2$
- Update q(^2) by computing A_{post} and B_{post}

3. Return m, S, A_{post}, B_{post}

Bayesian Gaussian mixture model

For homework.

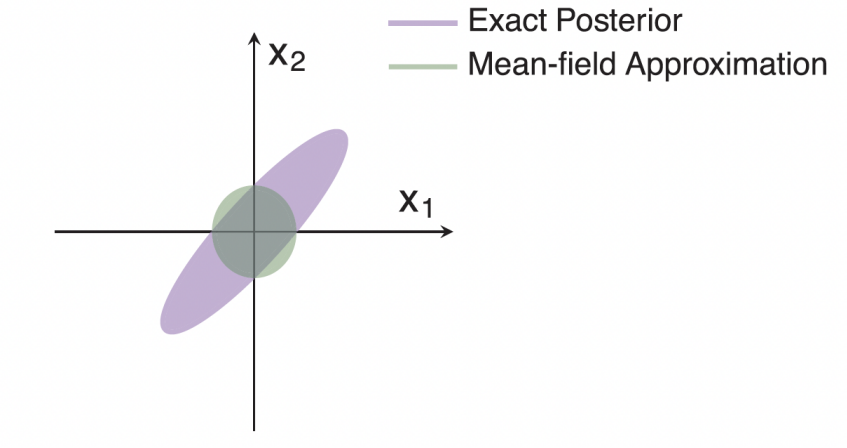
Variational inference with exponential families

In the example above we used exponential families for priors- this allows for easier derivation of CAVI updates that have a closed form.

There are a wide range of models that can use this approach:

- Matrix factorization, regression, stochastic block models of networks, and more
- For this class of models, generic updates for the exponential family distributions have been derived (see 4.1, Blei)

Drawbacks



Drawbacks

Exact posterior has positive correlation between x1 and x2. In other words, the posterior covariance matrix has non-zero off-diagonal entries.

- Mean field cannot capture correlation between variables
- Also underestimates marginal variances
- Means look good though

This is fine if all you want is point estimates! But if you care about uncertainty quantification, it's not that great.

Variational inference outside of mean field assumption

In principal, you can choose any q explicitly rather than have it fall out of the math:

- Pick a parametric family and optimize it's parameters (maximize the ELBO with respect to it's parameters using gradient ascent)
- Variational autoencoders
- Structured variational inference

Connection to the EM Algorithm

Let's say we have latent variables z, parameters θ, and data y. Fundamental problem is that the marginal likelihood of the data, $p(y|\theta) = \int_z p(y, z|\theta) dz$, is intractable.

- EM goal: maximize $p(y|\theta)$, the marginal likelihood
- VI goal: compute posterior $p(z, \theta|y)$

INLA

Performs approximate Bayesian inference in latent Gaussian models

- Also underestimate posterior variance
- Used often in spatial statistics

Lab

Implement Bayesian regression